EXP10

September 09, 2026

[ ]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, classification_report,␣
 ↪confusion_matrix
```

[ ]:
```
df = pd.read_csv(r"/content/Iris.csv")
print(df.head())
```

```
   Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm      Species
0   1            5.1           3.5            1.4           0.2  Iris-setosa
1   2            4.9           3.0            1.4           0.2  Iris-setosa
2   3            4.7           3.2            1.3           0.2  Iris-setosa
3   4            4.6           3.1            1.5           0.2  Iris-setosa
4   5            5.0           3.6            1.4           0.2  Iris-setosa
```

[ ]:
```
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Id             150 non-null    int64
 1   SepalLengthCm  150 non-null    float64
 2   SepalWidthCm   150 non-null    float64
 3   PetalLengthCm  150 non-null    float64
 4   PetalWidthCm   150 non-null    float64
 5   Species        150 non-null    object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
None
```

[ ]:
```
print(df.shape)
print(df.describe())
```

```
(150, 6)
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

[ ]:
```
if "Id" in df.columns:
  df = df.drop("Id", axis = 1)

df = df[df["Species"] != "Iris-virginica"].copy()
encoder = LabelEncoder()
df["species"] = encoder.fit_transform(df["Species"])

X = df[["SepalLengthCm", "SepalWidthCm"]].values
y = df["species"].values

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.2,␣
 ↪random_state = 42, stratify = y)

scaler = StandardScaler()
X = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

[ ]:
```
model = SVC(kernel = "linear", C = 1)
model.fit(X_train, y_train)

pred = model.predict(X_test)
```

[ ]:
```
print("Accuracy: ",accuracy_score(y_test, pred))
print("\nClassification Report:")
print(classification_report(y_test, pred, zero_division=0))
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, pred))

X_all = np.vstack((X_train, X_test))
y_all = np.concatenate((y_train, y_test))

x_min, x_max = X_all[:, 0].min() - 1, X_all[:, 0].max() + 1
y_min, y_max = X_all[:, 1].min() - 1, X_all[:, 1].max() + 1

xx, yy = np.meshgrid(np.arange(x_min, x_max, 0.02), np.arange(y_min, y_max, 0.
 ↪02))
```

```
Z = model.decision_function(np.c_[xx.ravel(), yy.ravel()])
Z = Z.reshape(xx.shape)
```

```
Accuracy:  0.5

Classification Report:
              precision    recall  f1-score   support

           0       0.50      1.00      0.67        10
           1       0.00      0.00      0.00        10

    accuracy                           0.50        20
   macro avg       0.25      0.50      0.33        20
weighted avg       0.25      0.50      0.33        20


Confusion Matrix:
[[10  0]
 [10  0]]
```